Building for Bushfire







Tasmania Fire Service

Planning and Building in Bushfire-Prone Areas for Owners and Builders

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fire.tas.gov.au

Tasmanian Government

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Bushfire is Tasmania's most prevalent natural hazard and a persistent threat to the safety of our communities, our built assets and other economic resources. Keeping our community safe from bushfires requires a range of measures in addition to firefighter intervention. Prevention measures, which include planning and building standards, provide the most cost-effective means of reducing risk to life and property.

The Tasmanian Government introduced significant reforms for planning and building in bushfire-prone areas in 2012. The reforms sought to establish consistent minimum standards for bushfire protection and to balance development in bushfire-prone areas with the physical, social and emotional risk of bushfire in our communities.

'Building for Bushfire' provides an introduction to planning and building requirements for bushfire protection in Tasmania. It is designed for those who are looking to buy or build in a bushfire-prone area and outlines the key issues that need to be considered.

The planning and building approvals process can be difficult and you should remember that your local council, accredited bushfire hazard practitioners and the Tasmania Fire Service are there to help.

Of course, planning and building for bushfire is just the first step in minimising the bushfire threat. It is also important to prepare your own Bushfire Survival Plan and to know your Community Protection Plan. The Tasmania Fire Service has good resources to help you prepare for an emergency.



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1 Bushfires in Tasmania

Bushfire is part of the history and natural landscape of Tasmania. Over the years, there have been many significant bushfires in Tasmania, including more recently the devastating fires in and around the Dunalley area early in 2013 that destroyed over 200 homes.

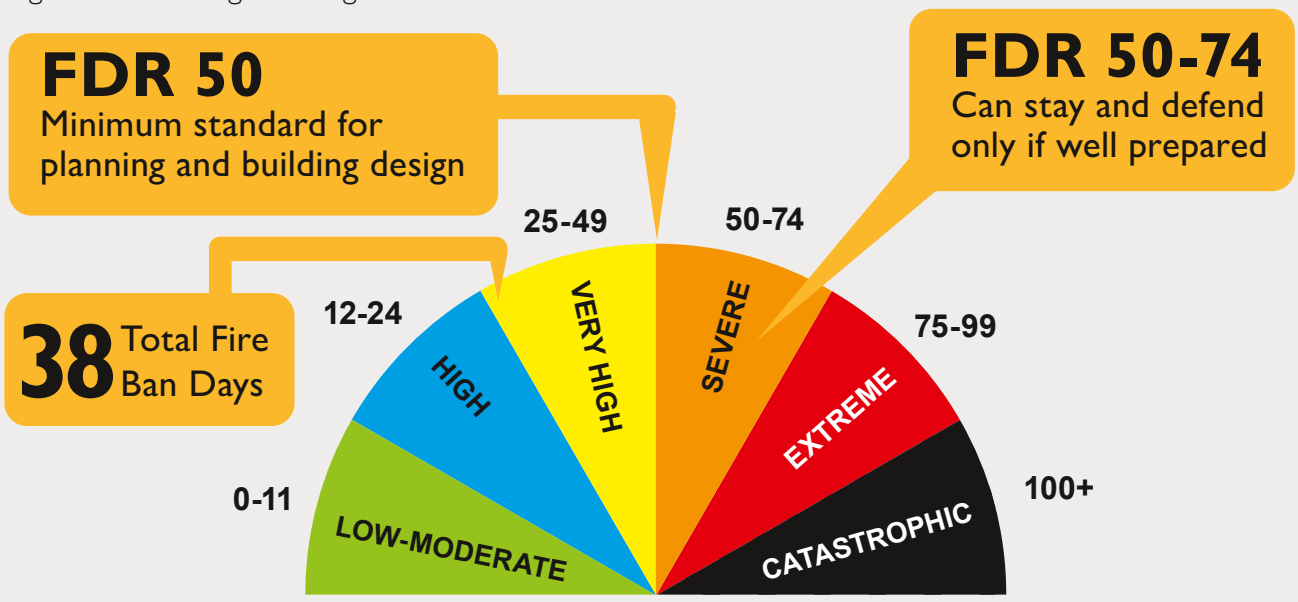
Where we settle, build our houses and construct our roads can greatly influence where a bushfire will eventually run. These factors play an important role in the inherent safety of an area. Good land-use planning and development can minimise the bushfire threat and reduce the physical, social and emotional costs of bushfire.

In October 2012, new requirements for developing in bushfire-prone areas were introduced in Tasmania to help reduce vulnerability to bushfire. These requirements incorporate lessons learned from bushfires within Tasmania and more broadly Australia, including the Victorian 'Black Saturday' fires of 2009.



The requirements mean that new development in bushfire-prone areas, is required to have a bushfire hazard management plan (unless an exemption can be obtained). If you are thinking of developing, building, or buying land in a bushfire-prone area, this Guide will help explain how these plans are created, the steps along the way, and where you can access help to work through the planning and building approvals process.

Figure 1: Fire Danger Rating



The Fire Danger Rating warns of the potential impact of a bushfire on any given day, based on forecast weather conditions. The Fire Danger Rating ranges from Low-Moderate (0) to Catastrophic (100+).

1.1 Building in a bushfire-prone area

If you are planning to develop in a bushfire-prone area, you need to plan to reduce your vulnerability to bushfire.

Tasmanian planning and building requirements aim to balance the benefits of development with the protection of life, property and environment during a bushfire. They also support the work of Community Protection Planning and any emergency response by the Tasmania Fire Service.

In Tasmania new development in a bushfire-prone area must be planned, designed and constructed to survive a fire with a Fire Danger Rating of 50 (see Figure 1). Your bushfire hazard management plan will detail how your development meets these minimum requirements.

It is important to remember, however, that even if your home conforms to the minimum requirements, bushfires can and will occur on days where the Fire Danger Rating exceeds 50, at which point homes may not be defendable. The Tasmania Fire Service will provide public warnings and advice on conditions during a bushfire, and whether you could defend your home or you should leave early.

The bushfire requirements include restrictions on the development of vulnerable (e.g. nursing homes) and hazardous uses in bushfire-prone areas. In some circumstances, a proposed use and development may not meet the minimum required standards, or the risks associated with the development may be too great.



1.2 Buying in a bushfire-prone area

If you are considering buying land or buildings in a bushfire-prone area, you should check with the local council to confirm whether the property is within a bushfire-prone area and if it has an existing bushfire hazard management plan.

A bushfire hazard management plan, if available, will provide useful information for you including how the land surrounding the building must be maintained and where you can build, renovate, and extend without needing to prepare a new plan. If a bushfire hazard management plan has been prepared for the property the owner must ensure the property remains compliant with all requirements of the plan. The property vendor or your local council should be able to advise if a bushfire hazard management plan exists for the property.

Many older properties may not have a bushfire hazard management plan. In these cases, a plan will only be required if you undertake new building work on the property. You should consider bushfire requirements if purchasing a property to develop.

It should be noted that under Tasmanian legislation a caravan or 'tiny home' that is permanently attached to the ground is considered to be a building and is therefore required to comply with building (including bushfire) requirements.

1.3 Rental property in a bushfire-prone area

If you intend to let a property you own, you must be aware of your legal obligations in regard to bushfire safety. Where a bushfire hazard management plan is in place, the property owner is responsible for ensuring compliance with the requirements of the plan, unless an alternate arrangement is explicitly detailed within a residential tenancy agreement. Maintaining compliance with a bushfire hazard management plan requires a range of routine maintenance tasks.

If you are renting property in a bushfire-prone area, you too need to know whether there are specific bushfire protection measures that you or the property owner are required to maintain. If you are new to the area, it is particularly important that you prepare a Bushfire Survival Plan so you are well prepared in the event of bushfire.

1.4 What's in this Guide

This Guide provides details on:

- What to consider in planning your development;
- Designing and building your development;
- Maintaining your property and responding in an emergency; and
- A checklist for planning and building in a bushfire-prone area.



2 Planning your development

2.1 Am I in a bushfire-prone area?

A 'bushfire-prone area' is an area where a bushfire is likely to occur that may cause significant damage to buildings and danger to life.

Some local councils have a planning scheme overlay that identifies bushfire-prone areas within their municipality. Planning scheme overlays can be viewed online through the LIST website (www.thelist.tas.gov.au).

In the absence of a published overlay, all land within 100m of 'bushfire-prone vegetation' is considered to be in a bushfire-prone area.

Your local council or an accredited bushfire hazard practitioners can advise you whether your land is bushfire-prone. Remember, it is better to check whether your land is bushfire-prone early in the planning process to save time and costs later (for more details, see *Section 7 - Further Information*).

If your land is bushfire-prone then your proposed development will likely require a bushfire hazard management plan.

What is bushfire-prone vegetation?

Bushfire-prone vegetation includes areas of vegetation larger than one hectare (ha). The one hectare area may be made up of clumps of vegetation separated by less than 20 metres, but will not include clumps of vegetation separated by more than 20 metres.

Bushfire-prone vegetation includes forests, shrubs and grasslands, but does not include maintained lawns, parks, orchards or horticultural land.

The role of accredited bushfire hazard practitioners

Bushfire hazard practitioners are accredited by the Tasmanian Fire Service to provide advice on matters relating to planning and building in bushfire-prone areas. In order to become accredited, bushfire hazard practitioners are trained in understanding how planning and construction relates in vulnerability to bushfires.

A list of accredited bushfire hazard practitioners can be found on the TFS website under 'Building for Bushfire' (www.fire.tas.gov.au).



2.2 What is a bushfire hazard management plan?

A bushfire hazard management plan is a plan that details the specific measures that will be implemented to achieve an acceptable level of protection for a proposed use or development.

Accredited bushfire hazard practitioners can prepare bushfire hazard management plans. They provide a report and a plan to demonstrate compliance with the applicable statutory requirements. An example of a plan is provided on the last page of this Guide.

Remember that not all development in Tasmania requires a bushfire hazard management plan, only development in bushfire-prone areas. Repairs, maintenance and minor extensions away from bushfire-prone vegetation may not require a plan.

Your bushfire hazard management plan will specify:

- The applicable Bushfire Attack Level (BAL) for construction;
- A building area with minimum setbacks;
- Standards for site access and firefighting water supplies;
- Hazard management areas and required vegetation removal;
- Ongoing maintenance requirements; and
- Other additional recommendations for risk mitigation as appropriate for the site.

2.3 When do I need a bushfire hazard management plan?

Depending on the zoning of your site, your new home may require a planning permit as well as a building permit.

For most types of building work (including residential dwellings), a bushfire hazard management plan is not required until the building approvals stage. This means you may not be required to provide this as part of your development application.

However, it is important to understand that bushfire requirements may have planning implications. For example, they may affect building siting and required vegetation removal. In some cases, bushfire requirements may even influence the choice of materials and design of the building. Therefore, bushfire requirements should be considered early in the design process and before submitting your development application. Engaging a bushfire hazard practitioner early in the design process is advisable. TFS recommends, wherever planning approval is required, a BHMP is provided with the application.

2.4 Balancing site location, construction requirements and maintenance

Deciding where to build on your land is an important step in planning for bushfire. The decision needs to balance how close your building will be to bushfire-prone vegetation, the slope of your land, how you will manage your land, and how access to the building and emergency water supplies will be provided.

A bushfire hazard management plan balances these considerations through an assessment of Bushfire Attack Level (BAL) for your development. Developments that are planned close to bushfire-prone vegetation are likely to have increased exposure to bushfire and will be assessed as having a high BAL. As a result, more stringent building construction standards will be required to improve resistance and reduce risk to occupants during a bushfire. In general, the higher the BAL the higher the cost of compliance.

The land between your development and bushfire-prone vegetation is part of your hazard management area and acts to reduce the impact of bushfires on your building; as such it needs to be maintained in accordance with the plan. Make sure you understand the costs associated with maintaining this area.

Figure 2: Bushfire Attack Level (BAL)

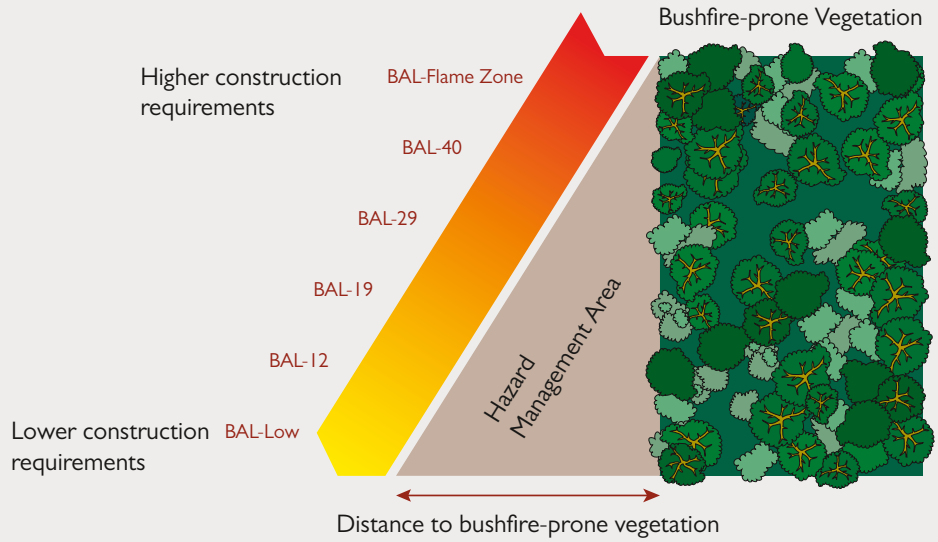


Figure 2 illustrates the relationship between Bushfire Attack Level and the distance to bushfire-prone vegetation.

Accredited bushfire hazard practitioners will be able to help you balance where you choose to site your building, additional construction requirements and the maintenance required over time. Remember that the location of your chosen building site will be shown on the bushfire hazard management plan. If you decide later on that you want to build in a different location, then a new or revised plan may be needed.

What is the Bushfire Attack Level?

The Bushfire Attack Level (BAL) is a measure of the potential impact of bushfire from ember attack, radiant heat and direct flame contact on your development. It takes into account the types of bushfire-prone vegetation surrounding your development, its proximity to your building site and the slope of the land.

The closer your development is to bushfire-prone vegetation, the greater the potential for bushfire to impact your building. As such, a high BAL may apply. The further your development is from bushfire-prone vegetation, the lower the potential for bushfire to impact your development and a lower BAL may apply.

There are six BAL levels for design: BAL-Low, BAL-12.5, BAL-19, BAL-29, BAL-40, and BAL-Flame Zone. Each level informs your building designer or builder of construction requirements to reduce the impact of a bushfire. The requirements for each BAL level are set out in Australian Standard 3959, *Construction of buildings in bushfire-prone areas.*

The applicable BAL for your development will be determined by your bushfire hazard practitioner.

2.5 Planning for access to buildings and water supply

In case of a bushfire, there must be safe access to and from buildings to the road network to support occupant evacuation and firefighter intervention.

Access must be designed and constructed so that:

- Occupants, fire fighters, and emergency services can get to and from the road network;
- Fire fighting vehicles and equipment can be manoeuvred and reach all parts of a habitable building; and
- Water supplies can be reached by fire fighting equipment.

In general, where a private access is constructed to buildings, it must have a minimum carriageway width of 4 metres and meet other engineering specifications.

Water supplies must be adequate, accessible and reliable for fire-fighting purposes. This can be achieved by siting the building within an acceptable distance of a fire hydrant (if available) or by installing a static water supply reserved for fire fighting purposes.



3 Building your new home

Once your bushfire hazard management plan has been prepared and planning approval obtained, the detailed design of your development can commence. Australian Standard 3959 *Construction of buildings in bushfire-prone areas* sets out construction requirements for new building work.

3.1 Construction requirements

The Standard contains six Bushfire Attack Levels that range from BAL-LOW (no additional construction requirements), through to BAL-Flame Zone (with the highest level of construction requirements). The Standard prescribes minimum construction standards designed to improve building resistance, as appropriate to the Bushfire Attack Level. This includes requirements for building features such as external walls, windows, doors, roofs, decks and water and gas supply pipes.

Requirements for ember protection include limiting gap widths in external facades and roof systems, screening openable portions of windows and hardening areas where embers and debris are likely to accumulate. This is particularly important as the majority of house losses from bushfire are the result of ember attack.



Resistance to radiant heat is achieved by reducing the combustibility of external elements and strengthening glazed elements appropriate to the Bushfire Attack Level. This has little effect on materials that can be used at BAL-12.5 and BAL-19 but becomes more restrictive at BAL-29 and above.

Building to the Standard has other benefits in addition to improving bushfire resistance:

- The use of durable products will increase the life of the building and reduce long-term maintenance costs;
- Minimising gaps in buildings means better weather-proofing and insulation properties;



- Higher specification glazing improves energy efficiency; and
- Sarking improves thermal performance of roofs and helps control condensation.

Your designer should clearly show how their design complies with the Standard.

3.2 Building Costs

The additional costs for a building project associated with building to Australian Standard 3959 depends on a number of factors including the type of construction proposed, the assessed Bushfire Attack Level and the other building requirements that apply.

Compliance costs will likely be more expensive for an elevated lightweight construction house when compared to a single storey, slab on ground house. The most costly aspects of compliance will likely be associated with gap and join sealing, upgraded glazing and screens.

Compliance costs need to be considered in the context of other aspects of the National Construction Code. For example, to comply with energy efficiency requirements many buildings will already require thermally efficient glazing and thermal mass.

A recent Western Australian study (Project BAL Build) indicates that compliance with Australian Standard AS 3959 increases overall building costs at BAL-12.5 by 3.8%, BAL-19 by 4.3% and BAL-29 by 7.7%. By comparison, building to BAL-40 and BAL-FZ was found to be much more costly, with up to 20.1% additional cost (noting that these higher BALs are rarely used in Tasmania) and are subject to additional regulatory requirements.

Case Study – Peter and Leah, a block on a new subdivision

Peter and Leah have just bought a block of land in a new subdivision. The developer provided the bushfire hazard management plan, which was created when the subdivision was planned. The plan shows the building area, the property's hazard management area and the Bushfire Attack Level, BAL-19, for their land.

Peter and Leah were happy with the plan, and talked to their designer about the construction features required to meet BAL-19. Some of the items discussed included:

- External cladding: Peter was free to choose weatherboard cladding and the designer incorporated bushfire-resisting boards near to the ground.
- Deck: Both Peter and Leah really wanted a timber deck. The designer ensured that non-combustible materials were incorporated within 300mm of the external walls.
- Windows and doors: Leah chose to install special bushfire screens across the windows and doors instead of bushfire shutters, which she didn't like the look of. The screens she chose were also cheaper.

There were no additional requirements for the foundations and roof. Gaps around the house were covered with mesh to prevent burning embers becoming caught in them. Their designer suggested a simple rectangular design as good practice to minimise the buildings' exposure to flames and heat.

Peter and Leah's bushfire hazard management plan also showed their hazard management area that had to be maintained to reduce the impact of bushfire on their home.

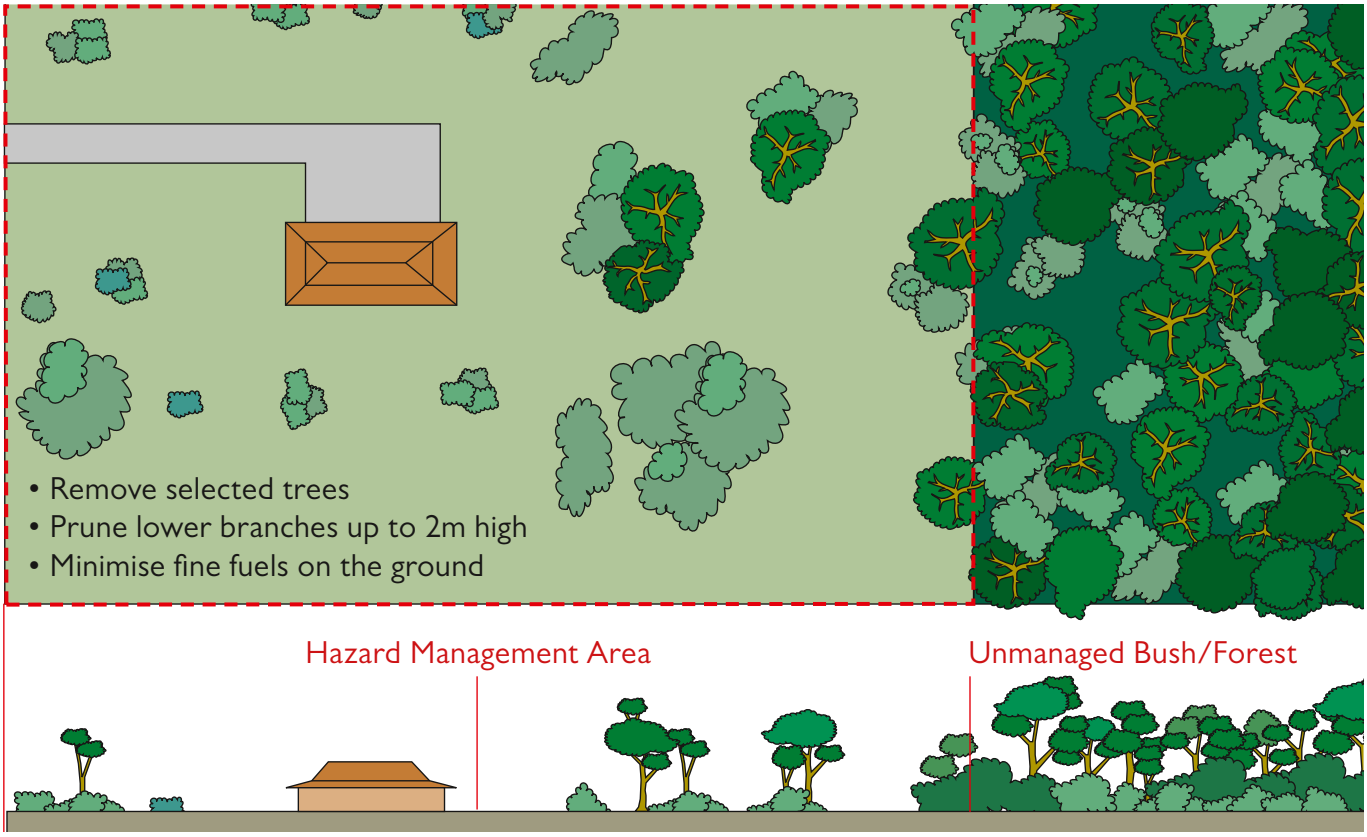


Figure 3: Hazard Management Area

4 Maintain your land

Your bushfire hazard management plan will specify an area around your building that is required to be kept in a low fuel state to prevent the spread of fire. Failure to maintain this area as specified can place buildings and lives at risk.

A hazard management area has two important roles. Firstly, it is much easier and safer to defend your home when most flammable material close to your building has been removed. Secondly, it reduces your building's exposure to radiant heat and flame contact. An example of a hazard management area is shown in Figure 3.

Your hazard management area must be maintained all year round. Some common maintenance includes:

- Remove selected small trees to maintain clumps separated by open areas, rather than continuous strips of vegetation;
- Prune lower branches and shrubs; and,
- Minimise fine fuels at ground level: mow, slash, and rake.



Note: While you should remove all trees next to your buildings, retention of some trees within your hazard management area helps to trap embers and reduce wind speeds around your building.



5 Prepare, act and survive

If you live in bushfire-prone areas then your home is at risk from bushfire. Most bushfires in Tasmania occur during relatively mild summer weather and are easily controlled by fire fighters. However, bushfires that break out on very hot, dry and windy days can spread rapidly and may be difficult or impossible for fire fighters to control. These fires can burn large areas of forest and farmland, destroy homes and livestock, and sometimes kill or injure people.

A properly prepared home is more likely to survive a bushfire than one that hasn't been prepared, and the chances increase significantly if able-bodied people are there to protect your home.

While properly prepared and defended homes can provide a safe haven during almost all bushfires, the Tasmania Fire Service recommends that in most cases, you should not plan to defend your home when the Fire Danger Rating exceeds 50 (severe) in your area.

The safest option always is to leave before a fire threatens your home. If you are not staying to defend your property, you need to plan to leave early.

Many people have died in bushfires because they have tried to relocate too late, and have been trapped and burnt in their cars or on foot. Fewer lives will be lost if people who choose to leave do so well before a bushfire threatens their home.

It is important to prepare your Bushfire Survival Plan, read your Community Protection Plan and know your Nearby Safer Place. For more information, visit fire.tas.gov.au



6 Checklist for building in bushfire-prone areas

There is a lot to think about when planning to build anywhere, let alone in a bushfire-prone area. The following checklist summarises the key issues for those looking to build in bushfire-prone areas.

Step 1: Is your land in a bushfire-prone area? (Section 2.1)

If you are developing land or building in a bushfire-prone area, you may need to consider bushfire as part of your development.

Talk to an accredited bushfire hazard practitioner or your Council to confirm if your property is in a bushfire-prone area.

Step 2: Plan and design (Sections 2.2 - 2.5)

Decisions made about where and how you develop your property will be informed by a variety of factors, not least of which is managing bushfire risk. Your bushfire hazard practitioner can help you to balance siting, construction requirements, vegetation removal and ongoing maintenance considerations.

Council will require a bushfire hazard management plan as part of the statutory approvals process.

Step 3: Build your new home (Section 3)

Once you have obtained the required permits you can develop your land or build your new home. Your building will be constructed to the Bushfire Attack Level specified on your bushfire hazard management plan. Any required hazard management areas will also need to be created prior to occupancy.

Step 4: Move in and Maintain (Section 4)

Congratulations! Now that you are living in your new home, your bushfire hazard management plan will show you how to maintain your hazard management area to protect against bushfire.

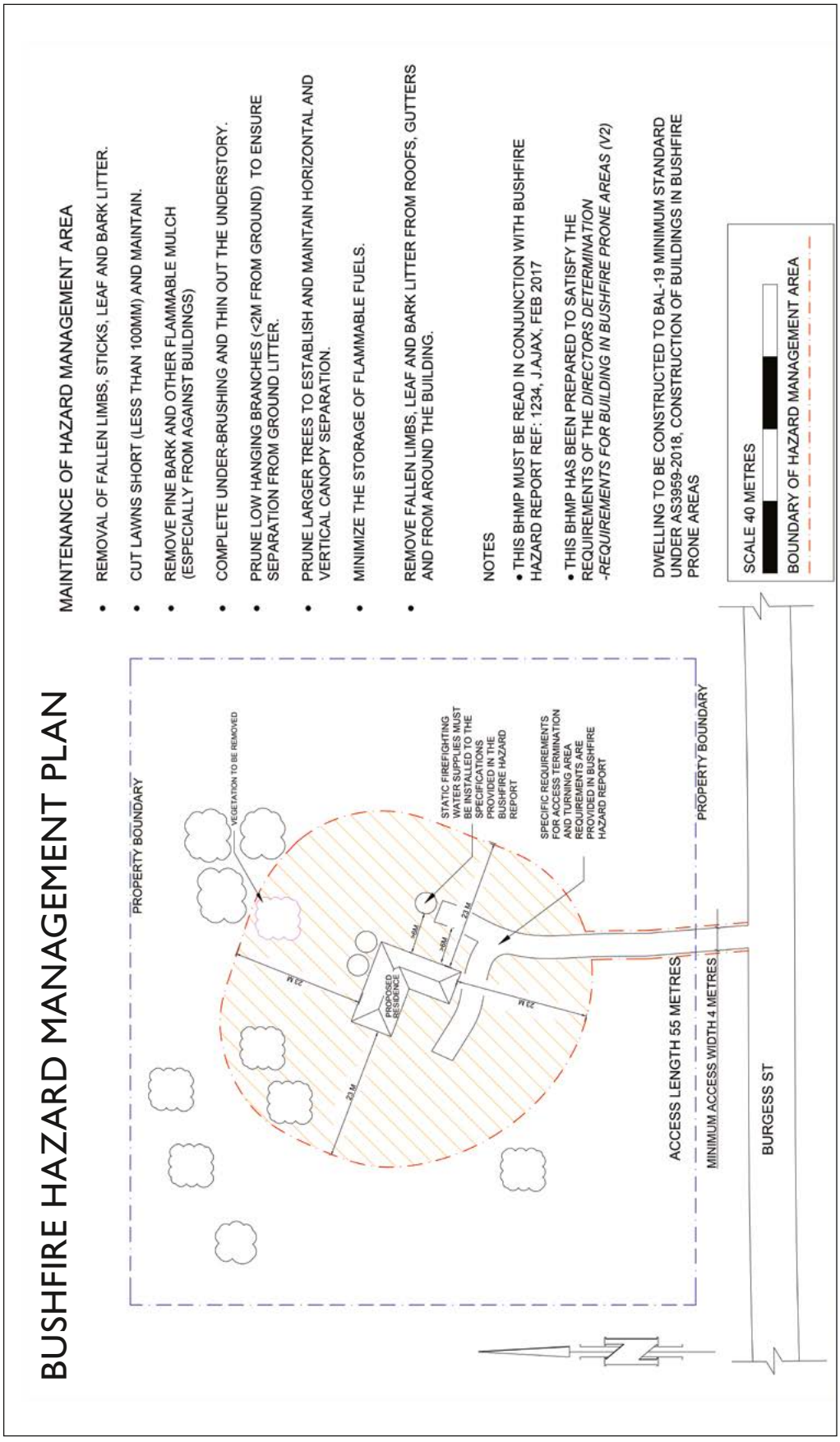
Step 5: Prepare for Emergency (Section 5)

Prepare your Bushfire Survival Plan. Know your Community Protection Plan and your Nearby Safer Place.

7 Further Information

TFS Online (www.fire.tas.gov.au)
Further information on bushfire requirements and a list of accredited practitioners can be found in the 'Building for Bushfire' section of the TFS website.

8 An example of a bushfire hazard management plan



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